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Heel Pain and Plantar Fasciitis: Hindfoot Conditions

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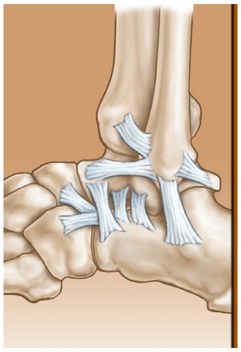

POSTERIOR HEEL PAIN

Pain in the posterior, superior portion of the calcaneus may be multifactorial, ranging from retrocalcaneal bursitis, enlargement of the superior bursal prominence of the calcaneus, insertional Achilles tendinosis, to inflammation of an adventitious bursa between the Achilles tendon and the skin (Fig. 120.1).[1-6] Each of these entities may exist as an isolated condition or may be part of a symptom complex. Careful analysis of the patient's subjective complaints and objective findings are required to arrive at the correct diagnosis. Disorders of the Achilles tendon are covered in Chapter 118 and are not reviewed in detail here.

Retrocalcaneal bursitis may occur as an isolated entity but is more commonly associated with the prominent posterior superior bursal portion of the calcaneus, or Haglund deformity. When Achilles tendinosis occurs concomitantly with this condition, it is generally located in the area of the Achilles tendon just at or above the insertion of the Achilles tendon at the posterior portion of the os calcis. Retrocalcaneal pain syndrome is commonly associated with a high-arched cavus foot and a varus heel.[1] The combination of these factors tends to produce a foot that does not dorsiflex as readily as a normal foot. The heel is prominent and is more susceptible to increased pressure from the tendons and the counter of the shoe.

History

The history is generally that of slow onset of dull aching pain in the retrocalcaneal area aggravated by activity and certain footwear. Footwear with a narrow heel counter is most commonly associated with this condition. Pain that starts after sitting or when arising from bed in the morning is commonly reported. At times, the patient may have a history of the acute onset of pain, which is sometimes associated with a traumatic incident. When this history is reported, a rupture of the Achilles tendon or disruption of calcific tendinosis must be considered.

Physical Examination

Physical examination reveals swelling in the area of the retrocalcaneal bursa between the Achilles tendon and the calcaneus.[3,4] A prominence is generally present in the area of the superior portion of the heel. The swelling in the retrocalcaneal bursa will be found just anterior to the Achilles tendon. By palpating medially and laterally at the same time, and with the aid of ballottement, one can sometimes feel fluid within the bursa (Fig. 120.2). With careful and discrete palpation, one can generally differentiate between swelling in the Achilles tendon and swelling in the retrocalcaneal bursa. The swelling of the Achilles tendon associated with retrocalcaneal bursitis is usually at the level of the tendon at or just proximal to the insertion. Dorsiflexion of the foot usually increases the pain in the area. A great deal of swelling and inflammation on examination may indicate involvement of both the retrocalcaneal bursa and the Achilles tendon. Redness and swelling may be present between the Achilles tendon and the skin, usually as a result of an adventitious bursitis produced by pressure of the shoe counter against the Achilles tendon. Periostitis may be present, which is a discrete localized area of tenderness of the os calcis, usually on the lateral side of the posterior portion of the os calcis and produced by pressure of the shoe counter. A "squeeze test," in which the palms of both hands apply moderate compression to the tuberosity, should be performed to rule out the presence of a stress fracture. If the patient reports pain with this examination, a high suspicion for a stress fracture should be noted.

Imaging

A lateral view of the foot is taken with the patient standing, which allows biomechanical evaluation of the foot and evaluation of the specific points of the os calcis. The points of the os calcis are identified as the posterior margin of the posterior facet, the superior bursal projection, the tuberosity indicating the site of the Achilles tendon insertion, the medial tubercle, and the anterior tubercle.[7] The shape and appearance of the superior bursal prominence are noted. Evaluation of the lateral radiograph may be performed using the method described by Fowler and Philip, which measures the posterior calcaneal angle (Fig. 120.3).[8] Fowler and Philip consider the bursal projection prominent if the angle is greater than 75 degrees. Some authors have concluded that a combination of the Fowler angle and the angle of calcaneal inclination is more effective in correlating the radiographic appearance with symptomatology than the Fowler and Philip angle alone, with the combined angle being greater than 90 degrees in patients with symptomatic Haglund disease.[9,10]

Parallel pitch lines have been used by Heneghan and Pavlov[3] to determine the prominence of the bursal projection (Fig. 120.4). The base line is constructed by placing a line along the medial tuberosity and the anterior tubercle, and a parallel line from the posterior lip of the talar articular facet. The bursal prominence is considered abnormal if it extends above this line.

Abstract

There are multiple etiologies for heel pain in the athlete, including retrocalcaneal bursitis and plantar fasciitis. As part of the initial workup, it is important to take a careful history, complete a thorough physical examination, and obtain appropriate imaging. Management may begin with conservative modalities, including physical therapy and orthotics. Pain that is refractory to nonoperative treatment can be addressed with surgical intervention.

Keywords

heel pain
plantar fasciitis
bursitis
retrocalcaneal

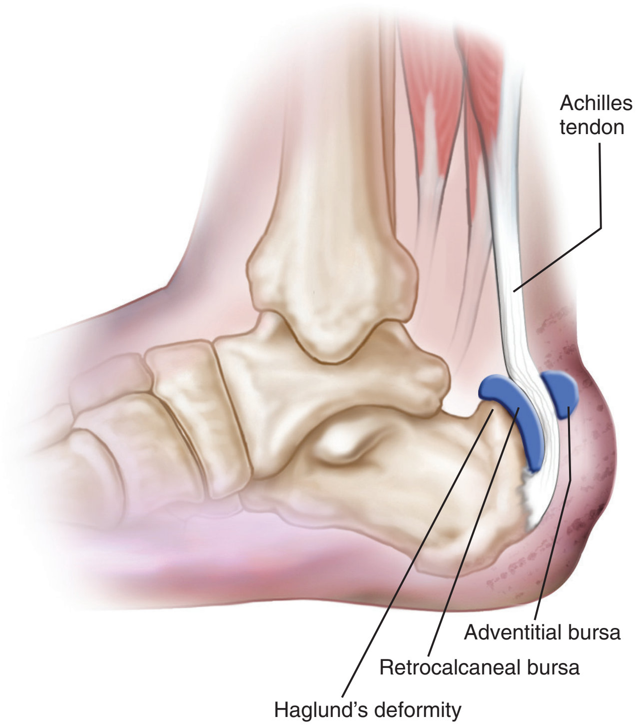

Fig. 120.1 Haglund deformity with a retrocalcaneal bursa between the Achilles tendon and the superior bursal prominence, and an adventitious bursa between the Achilles tendon and the skin.

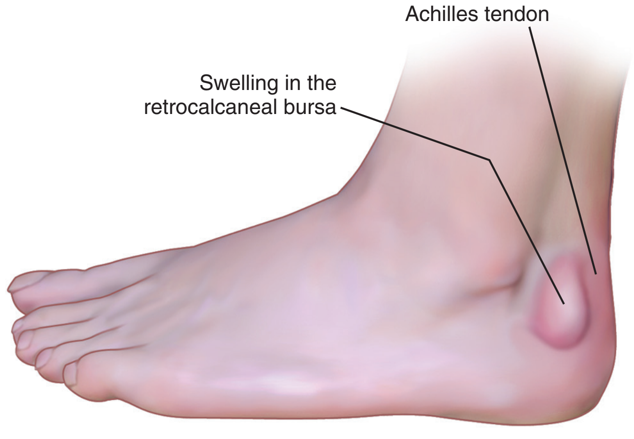

Fig. 120.2 The area of the swelling; retrocalcaneal bursitis with swelling is anterior to the Achilles tendon.

Magnetic resonance imaging (MRI) has provided clearer insight into the anatomic abnormalities associated with posterior heel pain.[11-14] The imaging allows visualization of the Achilles tendon and the bursa, as well as demonstrating any bony abnormalities in the posterior superior calcaneus. In patients with a suspected stress fracture of the calcaneus, an MRI scan can be ordered to delineate the diagnosis in an expeditious manner so that the most appropriate treatment may be initiated. In patients refractory to nonoperative treatment, a preoperative MRI will define which anatomic structures need to be addressed

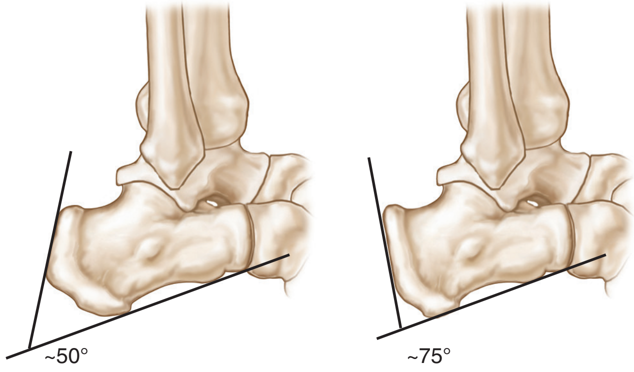

Fig. 120.3 Measurement of the Fowler and Philip angle. The normal angle is shown on the left, and an abnormal angle is shown on the right. The upper level of normal is considered to be 69 degrees. The drawing at the right indicates an abnormal angle of 75 degrees.

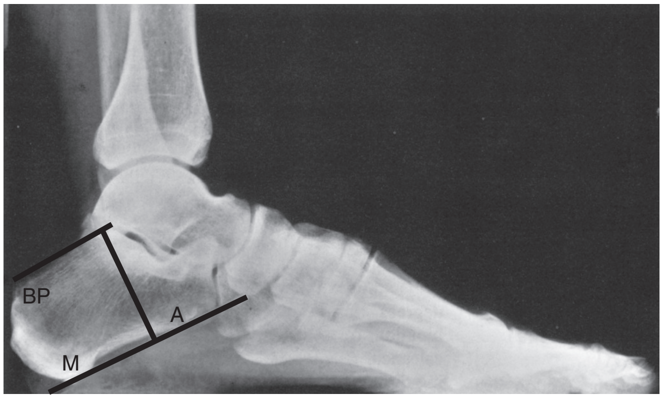

Fig. 120.4 Parallel pitch lines used to determine the prominence of the bursal projection. A line is drawn along the medial tuberosity *(M)* and the anterior tuberosity *(A)*. A parallel is constructed from the superior prominence of the posterior facet. If the bursal projection *(BP)* is above the superior line, the projection is considered abnormally large. (From Pavlov H, Heneghan MA, Hersh A, et al. The Haglund syndrome: initial and differential diagnosis. *Radiology* 144[1]:83–88, 1982.)

(Fig. 120.5). The degree of tendinosis present in the Achilles tendon is easily visualized and distinguished from isolated bursitis.

Decision-Making Principles

Nonoperative treatment is successful at alleviating posterior heel pain regardless of cause in most patients. Adventitious bursitis is generally treated conservatively by softening the heel counter with use of a small U-shaped pad to relieve the pressure of the shoe or counter against the inflamed area; in addition, anti-inflammatory medications are used. Surgical intervention performed solely for adventitious bursitis is unusual and carries a risk of wound slough, and thus it should be avoided if possible.

Retrocalcaneal bursitis and Haglund deformity are generally managed by conservative measures consisting of antiinflammatory medication, decreased activity, padding to prevent pressure on the affected area, orthoses or heel lifts, and strengthening and stretching exercises. If the condition does not respond to these modalities, then surgical intervention may be considered. Surgery generally consists of excision of the exostosis and the

Fig. 120.5 (A) Axial magnetic resonance imaging (MRI) of a normal Achilles tendon showing the normal shape of the Achilles tendon. (B) A sagittal MRI scan of a normal Achilles tendon showing the normal shape of the Achilles tendon. (C) A sagittal MRI scan showing increased signal in the insertion of the Achilles tendon consistent with tendinosis. Increased fluid is present, demonstrating an inflamed bursa surrounding Haglund deformity. (D and E), Axial and sagittal MRI scans demonstrating chronic tendinosis of the Achilles with marked fusiform swelling of the tendon.

retrocalcaneal bursa and, at times, the adventitious bursa, if present, along with correction of the Achilles tendon disease with tendon transfer if necessary. Although good results after surgery are reported in most series, in the athlete, this condition may present a serious threat to continued full activity, even after surgical intervention.

Treatment Options

Nonoperative Therapy

Initial goals of the treatment of patients with retrocalcaneal bursitis are to control pain and attempt to allow the patient to return to normal function and activity. Rest, particularly soon after the onset of symptoms, can be helpful. The duration of rest may be prolonged, depending on the increasing duration of symptoms.[15,16] Cross-training with low-impact exercises, such as an elliptical machine, swimming, or biking, may prevent deconditioning in the athlete. Modified regimens may be suggested for an athlete who is unwilling to cross-train.[16-18] In patients with exquisite tenderness, immobilization in a controlled ankle motion (CAM) boot or short-leg walking cast may be helpful.[17,19] Immobilization should be used cautiously in athletes, because patients can have resultant tendon and muscle atrophy, degeneration, and decreased blood supply.[18,20]

Nonsteroidal drugs that are administered orally or delivered locally in the form of a patch may decrease local inflammation.[20-22] Cryotherapy and ice can decrease pain, swelling, and inflammation as well.[20-22] Corticosteroids that are taken orally, injected locally, or applied topically have been used.[23] Injections must be used cautiously because they can lead to a higher incidence of tendon ruptures and tendinopathy.[24,25] Animal model studies with intratendinous injections have been shown to result in localized tendon necrosis and decreased mechanical strength. If used, the steroid injection must be placed anterior to the tendon, in the area of the retrocalcaneal bursa.[20,21] The patient is immobilized in a CAM walker for 3 to 4 weeks to minimize the risk of tendon rupture after injection into the retrocalcaneal bursa.

Orthoses may help these patients by providing a heel lift function, correcting hyperpronation, or minimizing leg length discrepancies.[20,21] Care must be exercised when correcting hindfoot pronation deformities, because overcorrection can result in inflexibility of the hindfoot with decreased shock absorption.[21] Heel cups can decrease the strain on an Achilles tendon and elevate a prominent superior calcaneal tuberosity away from the tendon. High heels, clogs, open-back shoes, gel braces, and horseshoe pads can also provide symptomatic relief for the patient.[19,26]

If retrocalcaneal bursitis is present with a normal Achilles tendon, conservative therapy is implemented until the patient has been asymptomatic for 4 to 6 weeks. The patient may then return to sports participation starting with limited activity and working up to full activity within 4 to 12 weeks, assuming that they have recovered full strength and mobility without pain. If retrocalcaneal bursitis is associated with degeneration of the Achilles tendon, nonoperative treatment should be used until the patient is asymptomatic; a gradual increase in activity is then allowed over a 6- to 12-week period.

Operative Therapy

If the patient does not respond to these modalities, then surgical intervention may be considered. Surgery generally consists of excision of the exostosis, the retrocalcaneal bursa, and at times the adventitious bursa, if present, along with correction of the Achilles tendon disease with tendon transfer if necessary. Although most authors report good results after surgery, in the athlete this condition may present a serious threat to continued full activity, even after surgical intervention.

Authors' Preferred Technique

Posterior Heel Pain

Surgical procedures are usually performed for retrocalcaneal bursitis associated with the superior bony prominence. The retrocalcaneal bursa and the superior bursal prominence are excised. The adventitious bursa is excised if it is prominent. If the adventitious bursa is excised, the surgeon must take care to excise it carefully and meticulously to avoid adversely affecting the blood supply or the skin overlying this area, because a skin slough can result in significant morbidity for the patient.

A medial or lateral incision, or a combination of incisions 1.5 cm anterior to the Achilles tendon, is made as determined by the location and width of the bony prominence (Fig. 120.6). Alternatively, a central posterior Achilles splitting approach can be used. Prone positioning can be used with all of the aforementioned incisions. A medial incision may be carried out with the patient supine and minimizes the risk of damage to the sural nerve, but the superior calcaneal exostosis is typically lateral, which may make excision slightly more difficult. Given the elimination of sural nerve injury and simplified positioning, this is the author's preferred technique in these cases. If the patient has concomitant tendinosis of the Achilles tendon, the central posterior tendon splitting approach is used.

After incision of the central aspect of the posterior skin, the incision is carried directly to the level of the paratenon, and dissection is performed at this level to maintain full-thickness skin flaps to avoid skin loss. Attention should be given to the calcaneal branch of the sural nerve on the lateral side and the medial calcaneal nerve on the medial side. The Achilles tendon is inspected to confirm the presence and degree of tendinosis. Regardless of the surgical approach, the retrocalcaneal bursa is now excised. An exostosectomy is performed, removing the bone from the area of insertion of the Achilles tendon to the superior portion of the posterior facet of the os calcis (Fig. 120.7). Adequate bone is removed, and the edges are smoothed with a rasp. If greater than 50% of the insertion of the Achilles tendon is elevated from the calcaneus, repair with suture anchors is recommended to avoid rupture. In the case of a concomitant tendinosis, a flexor hallucis longus transfer may be required if a significant portion (>50%) of the Achilles tendon is excised as described in Chapter 118. The split in the Achilles tendon is repaired with an absorbable monofilament number 0 suture. The subcutaneous tissue and skin are closed after the procedure. Compressive dressings and plaster splints are applied to maintain 15 degrees of ankle plantar flexion to maximize perfusion to the posterior skin.

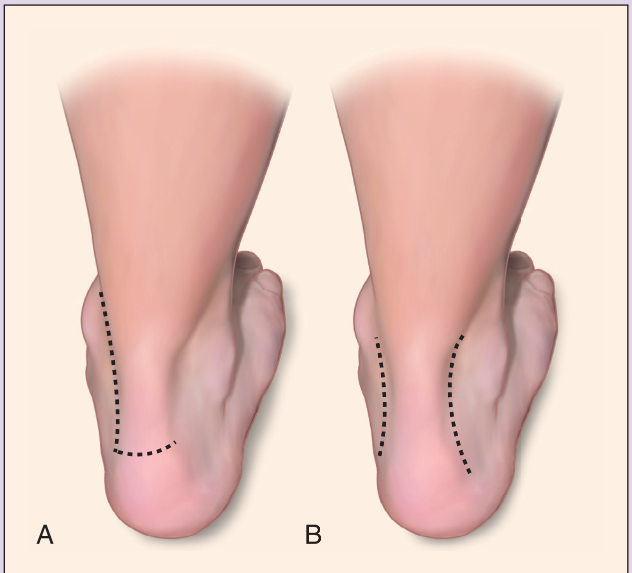

Fig. 120.6 Surgical incisions for retrocalcaneal bursitis. (A) The medial approach with J extension as described by Schepsis and Leach. (B) Medial and lateral approach incisions as described by Jones and James. ([A] From Schepsis AA, Leach RE. Surgical management of Achilles tendinitis. *Am J Sports Med.* 1987;15[4]:308–315 and [B] from Jones DC, James SL. Partial calcaneal osteotomy for retrocalcaneal bursitis. *Am J Sports Med.* 1984;12[1]:72–73.)

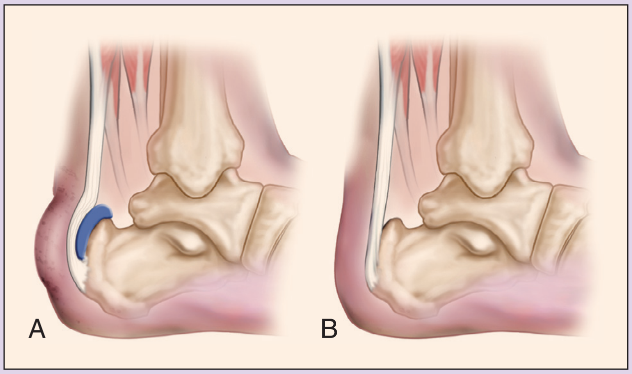

Fig. 120.7 (A) Haglund deformity with prominence of the posterior superior portion of the os calcis. (B) The appearance of os calcis after surgical resection of a posterior superior prominence for symptomatic Haglund deformity.

Postoperative Management

In cases in which the Achilles tendon is not involved, the patient is seen at 2 weeks for stitch removal, the lower extremity is placed into a removable CAM walker with the ankle at neutral position for an additional 4 weeks, and weight bearing is begun. A rehabilitation program for non-weight-bearing strengthening and range of motion (ROM) is begun at this initial visit. Six weeks after surgery, the walking boot is removed and the patient is allowed to weight bear with a $\frac{7}{16}$-inch heel lift during this time for 4 to 6 weeks. Therapy is continued and graduated to weight-bearing exercises. Patients are then advanced to regular footwear and continue to follow a strengthening program at home with a Theraband. Athletic activity is restricted for 3 months after surgery.

If surgery has been performed on the tendon in combination with excision of the retrocalcaneal bursa and exostosectomy, immobilization is continued for 8 weeks, but active (ROM) is begun at 3 weeks. Strengthening and stretching exercises are started at 6 to 9 weeks. Increased activity according to tolerance can be started at 12 weeks, and return to strenuous activity is allowed at 4 to 6 months if local symptoms have resolved.

Results

Ippolito and Ricciardi-Pollini[27] described three patients with invasive retrocalcaneal bursitis who had a large bursa and invasion of the os calcis. Pathologic examination revealed lymphoplasma cellular infiltrates containing proportionately more plasma cells than lymphocytes. Removal of the bursa provided clinical relief, and systemic rheumatic disease did not develop in later years.

Keck and Kelly[5] reported on 13 patients with 20 symptomatic heels that were treated surgically. In 17 heels, the superior bursa was excised, and in three heels a dorsally based closing wedge osteotomy was performed. Good results were reported for 15 of the heels treated. The initial results were good in all but two patients, whose pain recurred as a manifestation of generalized rheumatoid arthritis. An osteotomy was used to reduce the posterior prominence, and results were rated good in two heels—fair in one heel, and poor in two heels. The authors believed that too few osteotomies were performed in this series to evaluate this method. A disadvantage of the osteotomy was that a longer convalescence was required.[5]

A series of 65 patients with Haglund disease was reported by Ruch[9]; 17 underwent resection of the posterior superior portion of the os calcis, with resection of the posterior superior aspect both medially and laterally and removal of sufficient bone to render the previous palpable prominence entirely absent.[9] The patients were evaluated 6 months to 5 years after undergoing surgery. Fifteen demonstrated good to excellent results with elimination of symptoms. Three of the patients required a second procedure to obtain the desired result.

Schepsis and Leach[28] reported that the majority of athletes, particularly runners, who presented with acute or chronic posterior heel pain were successfully managed nonoperatively using a combination of (1) a decrease in or cessation of the usual weekly mileage, (2) temporary termination of interval training and workouts on hills, (3) a change from a harder bank surface to a softer surface, (4) a 0.25- to 0.50-inch lift inside the shoe or added to the shoe, and (5) a program designed to stretch and strengthen the gastrocnemius-soleus complex.[28] These measures were combined with use of oral anti-inflammatory medications and an occasional injection of corticosteroid into the retrocalcaneal bursa. Postural abnormalities were treated with orthotics. The authors retrospectively studied 45 cases of chronic posterior heel pain that were treated surgically in 37 patients. All but two of these patients were competitive long-distance runners who ran an average of 40 to 120 miles per week prior to the onset of symptoms. Their ages ranged from 19 to 56 years.

The surgical approach used by Schepsis and Leach[28] was a longitudinal incision 1 cm medial to the Achilles tendon that was continued transversely to form a J-shaped incision if necessary. A cast was applied and worn for 2 to 3 weeks, with weight bearing permitted after 1 week. When disease within the tendon required excision and repair, immobilization was continued for 1 to 2 weeks longer. (ROM) exercises were emphasized. A graduated program of swimming and stationary bicycling combined with isometric, isotonic, and isokinetic strengthening of the calf muscles was prescribed. Jogging was permitted after 8 to 12 weeks, but rarely sooner. Full return to a competitive level of sports activity usually required 5 to 6 months.

The patients were divided into three groups—those with Achilles tenosynovitis-tendinitis, those with retrocalcaneal bursitis, and those with a combination of both. In the 14 patients with retrocalcaneal bursitis, seven (50%) had excellent results, three (21%) had good results, and four (29%) had fair results. In the group with a combination of both conditions, five (71%) had excellent results and two (29%) had good results. It was noted that four of the six unsatisfactory results occurred in the group with retrocalcaneal bursitis.

Complications

More than half of all reported complications involve the surgical wound. Skin edge necrosis, superficial or deep infection, and seroma or hematoma formation are the most common complications. These complications can be minimized with postoperative splinting in 20 degrees of plantar flexion. Sural neuritis, tendon rupture, or disruption of the repair and deep vein thrombosis have also been reported. Failure to achieve complete relief of symptoms has been reported in up to 29% of patients with retrocalcaneal bursitis. Given the high demands of the athletic population, appropriate counseling regarding the outcomes is critical prior to performing surgery.

Future Considerations

Endoscopic techniques for the débridement of the retrocalcaneal space and the posterosuperior tuberosity of the calcaneus have been reported in the literature. Ortmann and McBryde[29] reported on 28 patients and 30 heels that were treated with endoscopic surgery; 86.67% of patients reported excellent results, with one major complication of an Achilles tendon rupture.[29] The medial column of the Achilles tendon, the sural nerve, and the plantaris tendons are at risk with use of this technique.[30] Despite these risks, the purported benefits of lower morbidity and recovery time make this procedure an excellent option for surgeons familiar with endoscopy.

PLANTAR FASCIITIS

In reviewing the literature, it is apparent that many different theories exist about the cause of subcalcaneal pain, and hence many different methods of treatment have been suggested.[31-43] It has been said that although this condition is familiar to all orthopaedic surgeons, it is probably fully understood by none.[36]

Snook and Chrisman[44] noted that conflicting literature exists on this subject with regard to two salient points: that there is no accepted explanation of the cause of the condition, and that no generally approved method of treatment exists. These investigators thought that the basic cause might lie in the subcalcaneal pad, which in some unknown manner lost its compressibility, either by local loss of fat with thinning or by rupture of the fibrous tissue septa. Ali[45] stated that "the painful heel is due to a fibrotic response, similar to plantar fibromatosis and not to the spur of bone, which is the end result of recurrent strain on the plantar fascia." Tanz[46] thought that "inferior heel pain is often due to irritation of a branch of the medial calcaneal nerve." Similarly, Baxter and Thigpen[31] and Przylucki and Jones[47] advanced the thought that the heel pain is due to an entrapment neuropathy that involves the branch of the lateral plantar nerve to the abductor digiti quinti. It has been noted that this branch passes more proximally than is shown in most anatomic studies and is in the area of the heel spur.[31,48]

Heel spurs may be present or absent, and may or may not be the primary pathologic entity in persons with heel pain. However, they must be considered in the context of the entire syndrome. Laboratory studies in most patients with subcalcaneal pain syndrome are normal. When the subcalcaneal pain syndrome is present, and especially if it is persistent and severe, consideration must be given to the diagnosis of a systemic disorder such as seronegative arthropathy. In patients with the subcalcaneal pain syndrome, it has been reported that the incidence of the subsequent development of a systemic arthritic disorder may be as high as 16%.[36]

History

The history usually reveals a slow but gradual onset of pain along the inside of the heel.[35-37] Occasionally the pain may be associated with a twisting injury of the foot, producing an abrupt onset of pain.[48] However, the clinical course is generally similar regardless of the onset. The location of the pain is generally described as along the medial side of the foot at the bottom of the heel. The pain is worse upon first arising in the morning and then decreases with increased activity. However, it may increase after prolonged activity. Periods of inactivity are generally followed by an increase in pain as activity is started again. Numbness of the foot is not present. When severe pain is present, the patient is unable to bear weight on the heel and will bear weight on the forepart of the foot.

Physical Examination

Specific examination of the foot reveals acute tenderness along the medial tuberosity of the os calcis along its plantar aspect. This tenderness may be at the origin of the central slip of the plantar fascia, or it may be deep, in which case it probably represents a deep inflammation, perhaps with involvement of the nerve to the abductor digiti quinti. The plantar fascia is palpated to determine whether the plantar fascia is tender just at its origin or throughout its course. The plantar fascia is also palpated for nodules, the presence of which suggests plantar fibromatosis. The plantar fascia is palpated both with the toes flexed so that it is supple and with the toes extended, which places tension on the plantar fascia. Careful examination of ankle ROM is essential to identify any concurrent gastrocnemius or soleus contractures.

Common conditions in the differential diagnosis of heel pain should be ruled out. Tarsal tunnel syndrome may generate pain in the distribution of the calcaneal branch and first branch of the lateral plantar nerve, and is typically elicited with palpation on the plantar medial aspect of the hindfoot, not directly under the foot. The associated sensory and motor changes are likewise not seen with plantar fasciitis (PF). Fat pad atrophy will often cause pain more posterior to the plantar tuberosity, directly under the calcaneus. Applying a squeeze around the calcaneus will elicit pain in patients with calcaneal stress fractures.

Imaging

Standing, full weight-bearing roentgenograms of the heel, including the foot, are taken in the anteroposterior and lateral standing projections. Roentgenograms taken in this manner will provide information about the osseous structures of the foot and specific details of the os calcis. Such radiographs will also help classify the foot as normal, flat, or cavus. The presence of a spur or calcification along the medial tuberosity will also be shown (Fig. 120.8). Axial non-weight-bearing views of the os calcis may be taken to provide information about the os calcis in a second plane.

Although much discussion has occurred about the relationship of the calcaneal heel spur to subcalcaneal pain, the relationship has not been definitely established. Tanz[46] stated that a heel spur is located in the origin of the short toe flexors and not in the plantar fascia. He noted that 15% of normal asymptomatic adult feet have subcalcaneal plantar spurs, whereas about 50% of adult feet with plantar heel pain have spurs. Tanz thought that heel spurs contributed to the plantar heel pain, although many patients with plantar heel pain did not have spurs. Snook and Chrisman[44] agreed with this supposition. In their report on 27 patients with subcalcaneal pain, they noted that 13 had a calcaneal spur and 11 did not have such a spur. Mann[49] stated

Fig. 120.8 An inferior plantar spur *(arrow)* commonly seen in patients with plantar fasciitis.

that "over a long period of time, proliferative bony changes at the origin of the fascia may lead to the formation of a spur."

Shmokler and colleagues[50] reviewed 1000 patients at random with radiographs of the foot. They found a 13.2% incidence of heel spurs. Only 39% of those with heel spurs (5.2% of the total sample) reported any history of subcalcaneal heel pain. Shmokler and coworkers[50] believed that these statistics tended to support the premise that the presence of a heel spur did not mandate pain.

Intenzo and colleagues[51] studied 15 patients reporting chronic heel pain who underwent three-phase technetium-99m methylene diphosphonate bone scintigraphy. Ten patients demonstrated abnormal scan findings consistent with PF, with uptake only in the early soft tissue phase, and they had responded to conventional therapy. Two patients were found to have calcaneal stress fractures, and one patient demonstrated a calcaneal spur that required no treatment. The remaining two patients had normal scans and did not appear to have PF clinically. These investigators found the three-phase bone scan useful in diagnosing PF and in distinguishing it from other causes of the painful heel syndrome.

Grasel et al.[52] evaluated various MRI signs of PF to determine if a difference exists in these findings between clinically typical and atypical patients with chronic symptoms resistant to conservative treatment.[52] These investigators found signs on MRI that included occult marrow edema and fascial tears. Patients with these manifestations seemed to respond to treatment in a manner similar to that of patients in whom MRI revealed more benign findings. MRI is not required to make the diagnosis of either PF or fibromatosis. The primary use is to rule out other conditions such as calcaneal stress fracture, plantar fascia rupture, neoplasm, and Baxter neuritis.

Decision-Making Principles

Operative treatment for PF is reserved for patients with moderate to severe symptoms who have not responded to at least 6 months (or more conservatively 12 months) of nonoperative management. Currently no prospective, randomized controlled trials exist that support any specific operative intervention for treatment of PF. Open techniques for plantar fascia release provide complete visualization of the fascia and allow the fascia to be incised in a controlled fashion. In addition, the first branch of the lateral plantar nerve can be decompressed with an open technique. Opponents of an open technique note a higher incidence of wound complications, infection, and postoperative pain, and a longer return to work. Endoscopic plantar fascia release for treatment of PF has been suggested to reduce complications and provide an earlier return to work and activity. However, endoscopic techniques are limited by a small field of view and may result in inadequate resection of the plantar fascia. In addition, the lateral plantar artery and first branch of the lateral plantar nerve is not exposed during this technique, and injury may occur.

In a patient with a tight gastrocnemius with a positive Silfverskiöld test, a gastrocnemius recession may relax the tension on the plantar fascia, relieving the pain. Successful relief of plantar foot pain after gastrocnemius recession has recently been demonstrated. Further research is required to validate these results.

Treatment Options

Nonoperative Therapy

Management of subcalcaneal heel pain should initially begin with nonoperative treatment. Although consensus does not exist on the efficacy of any particular conservative treatment regimen, it is agreed that nonsurgical treatment is ultimately effective in approximately 90% of patients. Because the natural history of PF has not been established, it is unclear how much of symptom resolution is in fact due to the wide variety of commonly used treatments. Activities are restricted according to the patient's symptomatic tolerance.

The mainstay of treatment is to stretch both the Achilles and plantar fascia. Simple plantar fascia stretching exercises are certainly the least expensive and easiest modality for patients to initiate. Plantar fascia–specific stretching has demonstrated superior success compared with isolated Achilles stretching (Fig. 120.9).

Nighttime dorsiflexion splints have demonstrated fair success in multiple randomized studies and are especially useful for relieving "start-up" pain in the morning. Resolution of symptoms has been reported as early as 12 weeks and may improve significantly even after 1 month of splint use.

Orthotics have demonstrated similar results to nighttime splints with better patient compliance. No conclusive evidence exists to support the use of a prefabricated versus a custom-molded orthotic. The use of a silicone heel pad is a simple and cost-effective method in patients without deformity.

Oral nonsteroidal antiinflammatory drugs have demonstrated trends toward improved patient pain and function but no statistically significant difference from placebo when used with other conservative modalities.

Localized administration of corticosteroids, either through injection or iontophoresis, has demonstrated reliable short-term relief but no significant long-term benefit. The risk of plantar fascia rupture or fat pad atrophy must be weighed against the limited short-term benefit of corticosteroid injections, and repeat

Fig. 120.9 A plantar fascia–specific stretch is performed with the leg in a figure four position. The contralateral hand is used to dorsiflex the hallux and ankle. The ipsilateral hand is used to palpate the plantar fascia to ensure that it is taut.

injections should be avoided. Iontophoresis may be a safer option but requires an increased time commitment and may be impractical in the typical clinic setting.

Cross-training activity is important. The patient should avoid repetitive impact activities such as running or use of a treadmill and instead cross-train with a bicycle or elliptical trainer. These options assist in maintaining conditioning and increasing flexibility while avoiding cyclic loading.

Once the patient has become asymptomatic without tenderness and has maintained this status for 4 to 6 weeks, a gradual increase in activity may be allowed. Use of the orthotic is continued for several months. After several months, use of the orthotic is discontinued unless the patient has a biomechanical abnormality of the foot, such as a flatfoot or cavus deformity. If the patient has a flatfoot deformity, a device designed to correct the biomechanical abnormality, support the foot, and prevent the abnormal biomechanical stresses along the plantar fascia and the medial side of the heel is used. If the patient has a cavus deformity, a soft orthotic designed to decrease the shock and increase the weight-bearing area may be used indefinitely, depending on the patient's symptoms. When a patient has a deformity and has had a significant episode of subcalcaneal pain requiring treatment, use of the orthotic can be continued permanently. Although the over-the-counter type of heel cup can be used initially to try to provide some symptomatic relief, in a patient with a true biomechanical abnormality, a specific orthosis should be used.

Postoperative Management

The patient is kept non-weight-bearing for 2 weeks and then can bear weight in a short-leg cast for 2 more weeks; increased activity is started at 12 weeks. This operation is used only for patients with recalcitrant conditions. It carries with it the expectation that the patient will probably, but not certainly, be able to return to his or her preinjury status.

Results

In 1986 Lutter[43] outlined the decision-making process in athletes with subcalcaneal pain. He described 182 patients with heel complaints related to sports injuries; most of the patients were runners (76%). Approximately 20% of these patients required 3 to 4 months of conservative treatment before returning to sports activity. Five percent had chronic heel pain and did not recover within 9 to 12 months. For these patients, a surgical approach was considered. The procedure used in these patients depended on the preoperative diagnosis and varied from release of the nerves to release of the fascia to complete exploration of the posterior tibial nerve and its branches, and release of the plantar fascia. Cycling or swimming was begun 2 weeks after the operation. Gentle walk-dash run training and a gradual escalation up to running was allowed approximately 6 weeks after surgery. Patients were asked to refrain from walking until they were pain free and had no tenderness. If pain occurred with increasing activity, the workup was cut by 50% until the patient could tolerate the workup without pain.

Shock wave therapy has been examined in the treatment of chronic PF in athletes who run. Rompe et al.[54] demonstrated that after 6 months of treatments three times a week with shock wave therapy, the treatment group experienced greater relief than the sham treatment group.

Baxter and Thigpen[31] performed 34 operative procedures in 26 patients with recalcitrant heel pain. The procedure consisted of isolated neurolysis of the nerves supplying the abductor digiti quinti muscle as it passed beneath the abductor with release of the deep fascia of the abductor hallucis longus and removal of

Authors' Preferred Technique

Plantar Fasciitis

In the athlete, the least amount of surgery commensurate with high likelihood of a good result should be performed. In the athlete with recalcitrant subcalcaneal pain syndrome who desires to continue athletic activity, the exact site of the abnormality is evaluated carefully by means of differential blocks using a long-acting anesthetic such as bupivacaine hydrochloride. This evaluation allows more precise localization of the exact area of the pathologic condition. If clear evidence exists of an isolated gastrocnemius contracture without evidence of tarsal tunnel or Baxter neuritis, an isolated gastrocnemius recession is considered.

Prior to surgery, nerve conduction and electromyographic studies are considered in patients in whom a tarsal tunnel syndrome is possible. Laboratory studies are performed to exclude systemic arthritis or spondyloarthropathies. Bone scans with technetium-99m may be considered if one suspects a fatigue fracture, or if the exact location of the pain is not clear.

For partial release of the plantar fascia, the patient is positioned supine with a small bump placed under the contralateral hip to increase the external rotation of the affected limb. A tourniquet should be used to minimize the bleeding to ensure appropriate identification of the neurovascular structures. In all cases, the tourniquet is released and appropriate hemostasis is obtained prior to closure to minimize the risk of a hematoma and irritation of the nerves that may compromise the end result. A medial oblique incision along the heel is made as described by Schon (Fig. 120.10).[53] Loop magnification may be used. The sensory branch of the medial calcaneal nerve is located, inspected, and preserved. If entrapment of the medial calcaneal nerve occurs as it comes through the fascia, this entrapment is explored and released. Subcutaneous fat is carefully dissected and retracted to expose the superficial fascia of the abductor hallucis, which is then incised. The abductor hallucis muscle is then retracted inferiorly, and the deep fascia of the muscle is released to decompress the first branch of the lateral plantar nerve. The abductor is then retracted superiorly to visualize the plantar fascia. The medial one third of the plantar fascia is released under direct visualization, and a 1-cm square segment is excised.

Any prominent calcaneal spur may be identified deep in the wound and carefully débrided back to a smooth, low-profile base. Removal of the spur is not required and may increase the risk of plantar fascia rupture and injury to the lateral plantar nerve. No specific criteria exist on which to base excision of the spur; only that "large" spurs should be excised.

After copious irrigation and hemostasis, the incision is closed with a nylon mattress suture and a short-leg cast is applied. The patient should not bear weight for 4 to 7 days. Weight bearing in a cast or an over-the-counter brace is maintained for 14 more days. At 3 weeks, weight bearing with a shoe is permitted. Running is started at 6 to 12 weeks, and activity is then allowed as tolerated. If the patient has a biomechanical foot abnormality, an orthotic device is used after the operation.

Continued

Authors' Preferred Technique

Plantar Fasciitis—cont'd

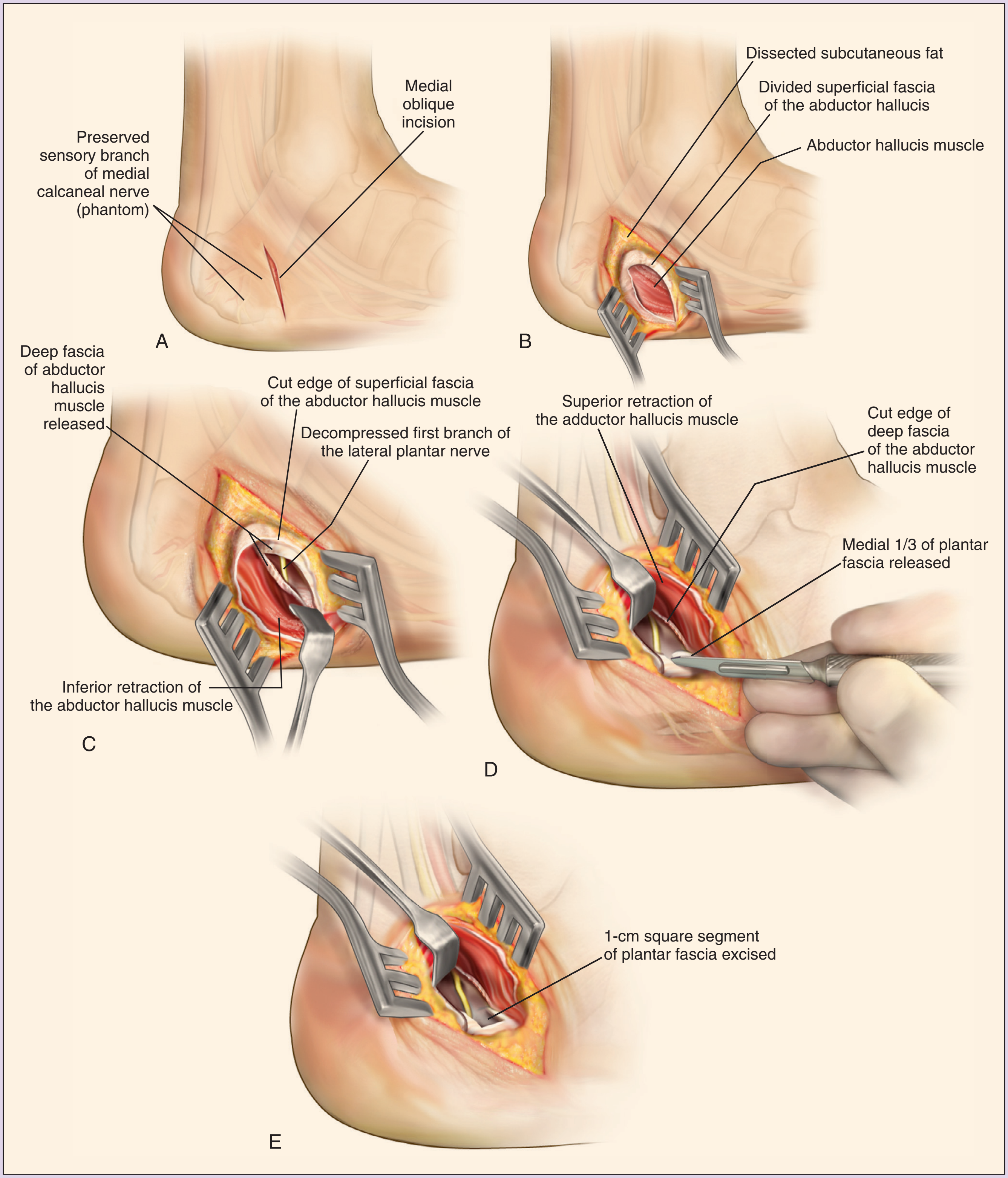

Fig. 120.10 Plantar fascia and nerve release. (A) Incision. (B) Release of the abductor hallucis muscle. (C) The abductor hallucis muscle is reflected proximally. (D) The abductor hallucis is retracted distally. (E) Resection of a small medial portion of the plantar fascia.

the heel spur if it impinged on or produced entrapment of the nerve.[31] Among the 34 heels operated on, 32 had good results and 2 had poor results.

Clancy[39] treated patients with a medial heel wedge and flexible leather support, heel cord stretching, and rest for 6 to 12 weeks, with a gradual return to running while wearing the orthotic and the medial heel wedge for 10 weeks. In patients who failed to respond to this treatment, surgery consisting of release of the plantar fascia and the fascia over the abductor hallucis longus was recommended. The 15 patients in whom surgery was performed returned to running within 8 to 10 weeks.

Henricson and Westlin[55] described 11 heels in 10 athletes who had chronic heel pain for which conservative therapy did not provide relief. The pain was due to compression of the calcaneal branches of the tibial nerve. Entrapment of the anterior calcaneal branch occurred where the nerve passed between the tight and rigid edges of the deep fascia of the abductor hallucis and the medial edge of the os calcis. Surgery consisted of identifying and releasing the tibial nerve and both calcaneal branches, and releasing the deep fascia of the abductor hallucis. Follow-up for 58 months after surgery revealed that 10 of the 11 heels were asymptomatic. The patients had resumed athletic participation after an average of 5 weeks.

Leach and coworkers[56] described 15 competitive athletes in whom 16 operations were performed. Surgery consisted of release of the plantar fascia at the insertion of the os calcis, making the incision along the medial aspect of the heel. In one instance, the medial calcaneal nerve was involved in the inflammatory process. One patient returned to running at 6 weeks; the majority returned to running 9 weeks after surgery. Most patients continued to improve up to 6 months after the surgical procedure. Of the 15 operations, 14 were entirely successful in that the athletes returned to their previous level of activity. One failure occurred in a marathon runner who improved but was unable to train at the level he desired. No complications were reported.

In 1984 McBryde[57] reported that in his running clinic, PF comprised 9% of the total running disorders seen. The conservative (nonoperative) approach of McBryde and his group consisted of (1) ice massage for 2 minutes four to six times daily, including before and after runs, (2) heel cord stretching for 3 to 5 minutes three to four times daily, (3) posterior tibial and peroneal strengthening, (4) heel cushioning and control, and (5) use of antiinflammatory medication. This regimen was usually successful in treating runners with PF who were seen within the first 8 weeks. In runners with symptoms lasting longer than 6 weeks, a period of absolute rest with use of a cast was usually required. Orthoses were used. Five percent of the patients in the series underwent surgery, consisting of plantar fascial release through a short 1-inch longitudinal incision in the medial arch. All returned to a successful running program 6 to 12 weeks after surgery. Overall, among the 100 patients with PF, 82 recovered with a conservative approach, 11 stopped running, 5 underwent surgery (all of whom returned to running), and 2 refused surgery and continued to be symptomatic.

Snider and colleagues[58] reported 11 operations for plantar fascial release for chronic fasciitis in 9 distance runners who had had symptoms for an average of 20 months and had not responded to nonsurgical treatment. The results of the operations were excellent in 10 feet and good in 1 foot, with an average follow-up time of 25 months. Eight of nine patients returned to their desired level of full training at an average time of 4.5 months.

Rask[59] reported a medial plantar neurapraxia that he termed *jogger's foot*. Three cases were reported in which there was probable entrapment of the medial plantar nerve behind the navicular tuberosity in the fibromuscular tunnel formed by the abductor hallucis; the inciting factor was eversion of the foot. All three patients were treated successfully with conservative measures, including a change in the running posture of the foot, antiinflammatory medication, and proper footwear.

In 2004, Saxena[60] reported on 16 athletes with intractable heel pain for whom conservative care had failed. Most of these athletes were runners. These patients were treated surgically with a uniportal endoscopic plantar fasciotomy. Saxena found that runners were able to return to athletic activity on average 2.6 months after surgery. Five poor results were found in patients with a body mass index greater than 27.

Complications

Acute complications after plantar fascia surgery are rare. Although few case series exist to determine reliable complication rates, the most commonly reported include wound dehiscence or infection. The most common complications reported after operative treatment for PF include further tearing or injury to the plantar fascia, collapse of the medial longitudinal arch, iatrogenic injury to the first branch of the lateral plantar nerve or neuroma formation, complex regional pain syndrome, and persistent pain. Complete excision of the plantar fascia has been associated with lateral column syndrome that presents a pain with weight bearing focused at the lateral border of the foot along the calcaneocuboid joint and fourth/fifth tarsometatarsal joints. In addition, recurrence rates after subtotal plantar fasciectomy have been reported as being approximately 10% in multiple studies.

Future Considerations

Increased focus on the role of a tight gastrocnemius complex and the effect on the function of the foot have begun to alter the surgical treatment for PF. With further research on biomechanical and clinical effect of a gastrocnemius release in patients with PF, the need for direct release of the plantar fascia may decrease. Although neither operation can be considered benign, patients with lateral column syndrome or injury to the lateral plantar nerve are severely limited without a good surgical recourse. Although injury of the sural nerve may occur, resection and burial can be performed, yielding an adequate result that does not preclude sporting activity.

For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:

Yu J, Park D, Lee G. Effect of eccentric strengthening on pain, muscle strength, endurance, and functional fitness factors in male patients with achilles tendinopathy. *Am J Phys Med Rehabil.* 2013;92(1):68–76.

Level of Evidence:
II

Summary:
The authors describe a prospective comparative study of 32 patients with Achilles tendinopathy who were placed into two treatment groups. One group underwent concentric strengthening and one underwent eccentric strengthening for an 8-week period. The authors were able to demonstrate a significant improvement in pain, balance, agility, and endurance in the patients who underwent eccentric strengthening. This article reinforces the critical nature of ensuring that therapy instructions are appropriately written for this disease process to focus on eccentric strengthening.

Citation:
Kearney R, Costa ML. Insertional achilles tendinopathy management: a systematic review. *Foot Ankle Int*. 2010;31(8):689–694.

Level of Evidence:
III

Summary:
The authors provide a systematic review of the literature regarding both operative and nonoperative management of insertional Achilles tendinopathy. Significant evidence favored conservative management, including both eccentric strengthening and shock wave therapy. The paucity of quality literature regarding operative intervention resulted in an inconclusive decision on the efficacy of surgery.

Citation:
Digiovanni BF, Nawoczenski DA, Malay DP, et al. Plantar fascia-specific stretching exercise improves outcomes in patients with chronic plantar fasciitis. A prospective clinical trial with two-year follow-up. *J Bone Joint Surg Am*. 2006;88(8):1775–1781.

Level of Evidence:
II

Summary:
The authors of this very significant article definitively demonstrate that plantar fascia–specific stretching is the critical component in the nonoperative treatment of plantar fasciitis. Efficacy was noted 2 years after the initiation of treatment with a 92% satisfaction rate.

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